

```
1  library ieee;
2  use ieee.std_logic_1164.all;
3
4  entity eight_to_fourEncoder is
5      port(s: in std_logic_vector(1 downto 0);
6           input1, input0: in std_logic_vector(3 downto 0);
7           en: in std_logic;
8           output8_4_en: out std_logic_vector(3 downto 0));
9  end;
10
11 architecture synth of eight_to_fourEncoder is
12 begin
13     process (s, en) begin
14         if s(1) = '1' and en = '1' then
15             output8_4_en(3 downto 0) <= input1(3 downto 0);
16         elsif s(0) = '1' and en = '1' then
17             output8_4_en(3 downto 0) <= input0(3 downto 0);
18         end if;
19     end process;
20 end;
```